

Sparse-sensing state estimation of vortex-gust airfoil interactions

Toward model-based flow control

A. Solera-Rico, A. Miró, O. Lehmkuhl, **C. Sanmiguel Vila**

csanvil@inta.es



uc3m

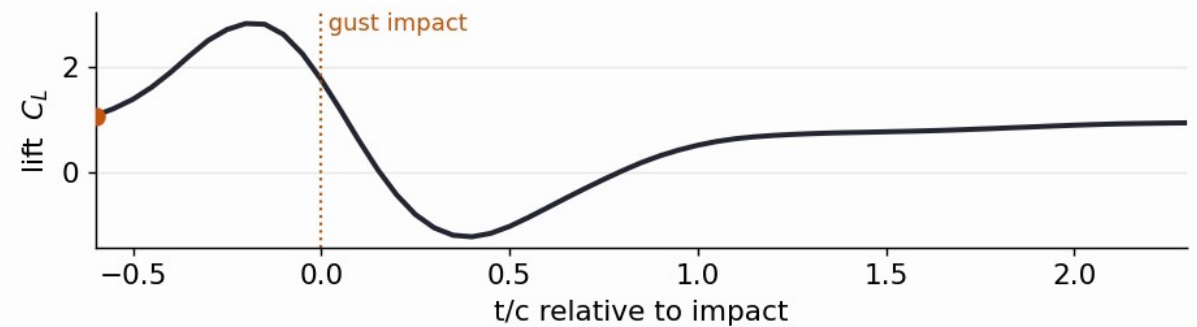
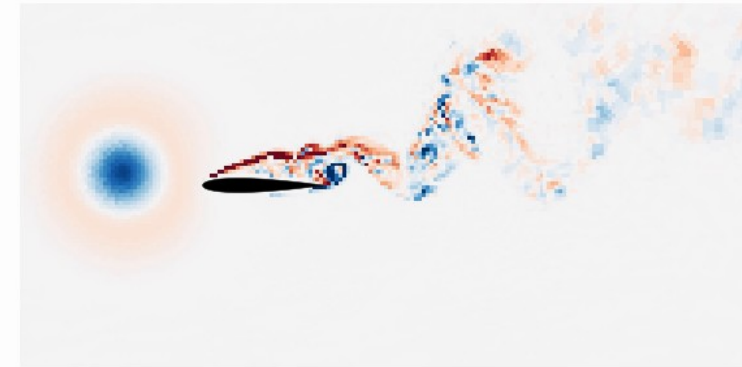
Universidad
Carlos III
de Madrid



The control problem: gusts are fast, strong, nonlinear

- Small and micro air vehicles increasingly fly where the **gust speed is comparable to or above the flight speed** (urban canyons, building and ship wakes): gust ratio $G > 1$ brings large, fast load transients classical models miss
- A vortex strikes the wing: a **leading-edge vortex** forms and sheds, and the lift swings strongly (right: DNS, a moderate $G = -2$ gust, C_L from about -1 to +3)
- Active control, model-based or RL, needs a **fast, faithful forward model** it can trust under rollout
- **This talk: what reduced state is worth planning against?**

Gust encounter ($G=-2.0$, $D=0.5$, $Y=+0.2$): $t/c = -0.60$



Objective: a reduced state that keeps the wake

- The load transient is built by the **leading-edge vortex (LEV)** and the wake it leaves: the discriminating information is in the **wake**, not the integrated forces C_L , C_D
- We quantify it by the **large-scale wake enstrophy**: a Gaussian scale split (Motoori & Goto 2019) at $\sigma/c = 0.05$ isolates the load-bearing large-scale vorticity (LEV + shear layer) from fine turbulence, then integrates ω_L^2 over the wake
- **Objective**: find a reduced state that faithfully **encodes the wake** (the part reconstruction loses), not just the forces, and that is **recoverable from sparse sensors**
- And **compare candidate states**, predictive vs reconstructive vs linear, under one matched protocol

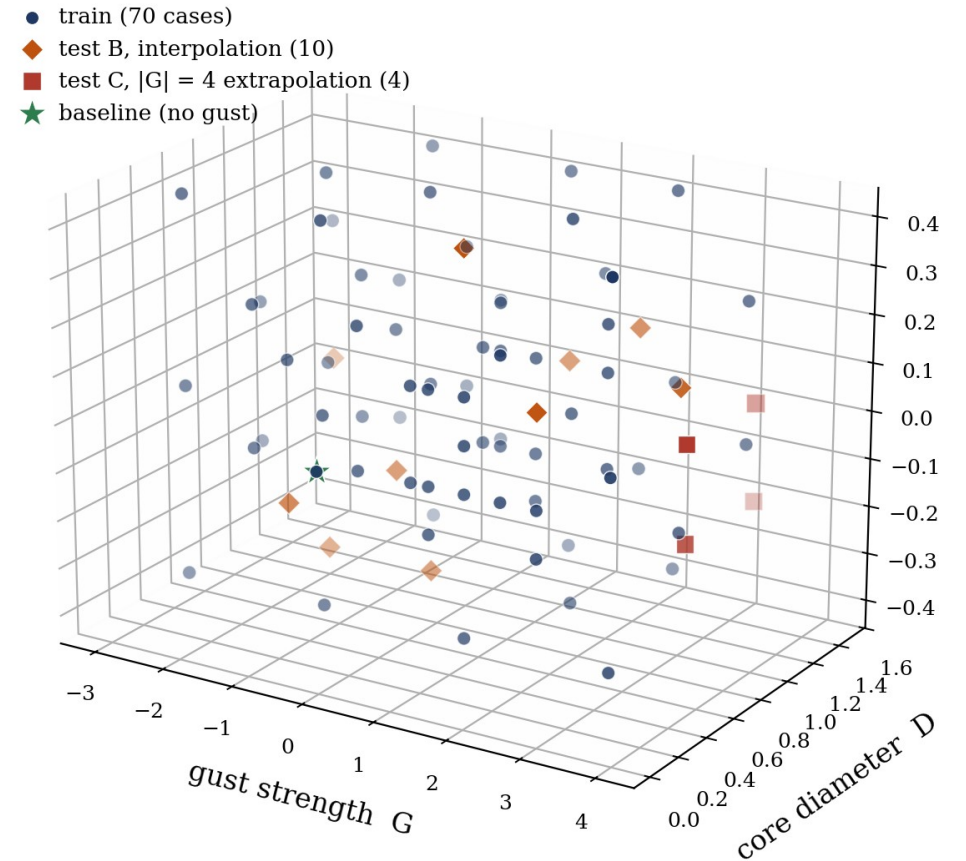
The opening for JEPA

- Most flow ROMs, POD/DMD, autoencoders, are trained to **reconstruct the field**
 - reconstruction fixes the latent only up to a diffeomorphism, so its geometry is not constrained to be predictable
- Alternative: learn a state that is **predictive by construction**, and plan against it

Which reduced state faithfully encodes the wake that reconstruction loses, and is recoverable from sparse sensors?

Data

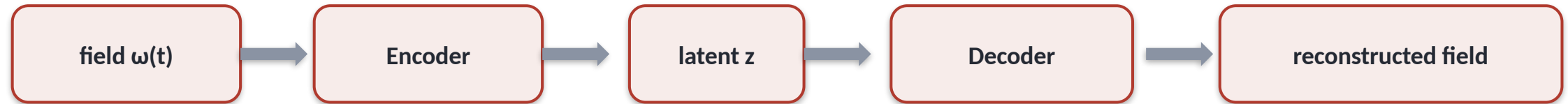
- **DNS** (SOD2D, no subgrid-scale model), NACA 0012, $\alpha = 14^\circ$, **Re** = 5000
- Taylor-vortex gusts parametrised by strength **G**, core diameter **D**, wall-normal offset **Y** (right: the 84-case envelope, isometric)
- Impact-centred encounters; five observables: C_L , C_D , **wake enstrophy**, $\pm$ circulation
- Held-out **test_b** (interpolation) and **test_c** ($|G| = 4$ extrapolation)



Training envelope in (G, D, Y) : train, test_b interpolation, test_c $|G| = 4$ extrapolation, baseline.

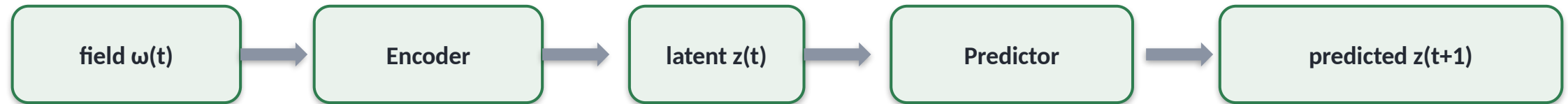
What is a JEPA? (1 / 2)

Autoencoder (most flow ROMs): reconstruct the field



loss compares the reconstruction to the input in PIXEL space

JEPA: predict the next latent, no decoder



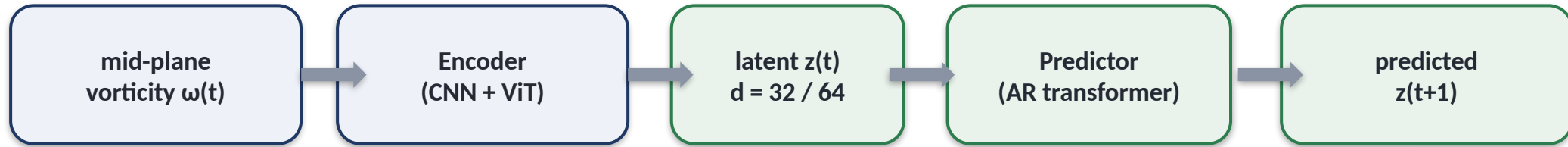
loss compares predicted $z(t+1)$ to $z(t+1) = \text{Encoder}(\text{next frame})$ in LATENT space; an anti-collapse term replaces the decoder

Reconstruction ties the latent to pixels; prediction makes it forecastable, which is what a controller needs.

What is a JEPA? (2 / 2): the recipe and why for control

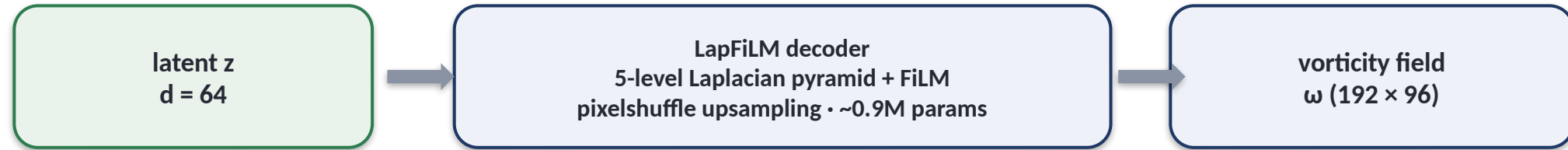
- **Training, in one line:** minimise the **latent** prediction error plus an **anti-collapse term** so the encoder cannot cheat by mapping everything to a constant (no decoder, no pixel loss)
- So the encoder **keeps only what predicts the future** (the control-relevant dynamics), and the predictor **gets no pixel-level shortcuts**
- **Cheap to roll out:** planning happens in the $d = 32-64$ latent, two to three orders of magnitude cheaper than evolving the full field
- Lineage: I-JEPA / V-JEPA (image, video); from-pixels LeWM, LeJEPA, PLDM, so far only gridworld and toy visual tasks
- **Why for control:** a learned world-model that is **observable from sensors**; to our knowledge the **first end-to-end JEPA on a parametric flow**

Our architecture for gust encounters



- The **encoder** maps the field to a latent: a pure state map $\omega(t) \rightarrow z$
- The **predictor** advances the latent $z(t) \rightarrow z(t+1)$ from the latent history; it **is** the latent dynamics model you plan or learn against
- A **visualisation decoder** is trained on the **frozen** encoder, never in the loss (figures only)
- Compared at matched latent dimension against a reconstructive AE (Fukami) and POD

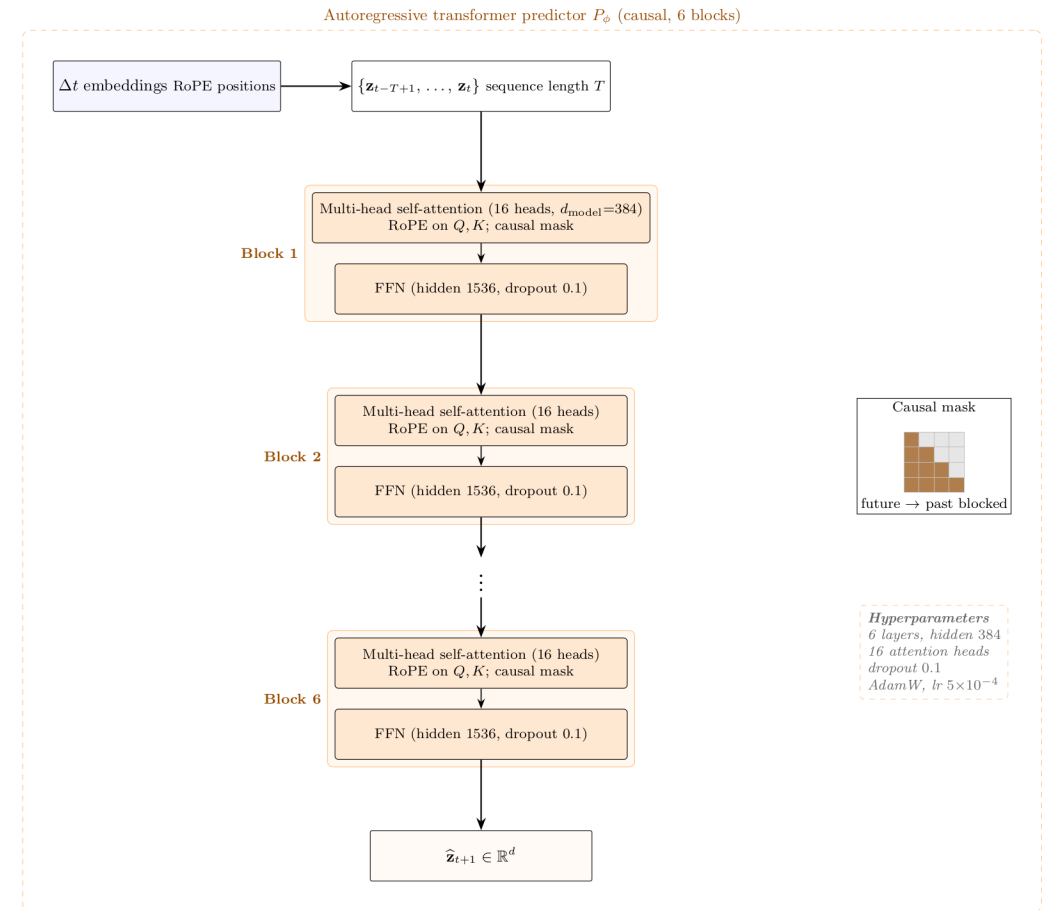
The visualisation decoder (figures only, never in the loss)



- Trained on the **frozen** encoder, **never in the JEPa loss**: it only renders a latent as a field so we can look at it
- **Why not a plain CNN or ViT decoder?** A transposed-conv CNN under a pixel objective **blurs the wake** and leaves checkerboard artefacts; a ViT decoder is heavier and patch-blocky. The **Laplacian pyramid** builds the field coarse-to-fine (large-scale wake first), **FiLM** lets the 64-D latent steer every scale at just ~0.9M params, and **pixelshuffle** upsamples without checkerboards
- **Loss** = region + Laplacian-pyramid + gradient + **spectral-amplitude** (small enstrophy / circulation terms): it matches structure **across scales and in the Fourier spectrum**, not pixelwise
- **Why SSIM, not MSE?** MSE is pointwise and **rewards blur** (a smoothed field scores well but loses the vortices); **SSIM** compares local contrast and structure, crediting a correct wake **pattern**. Held-out **SSIM ≈ 0.73**

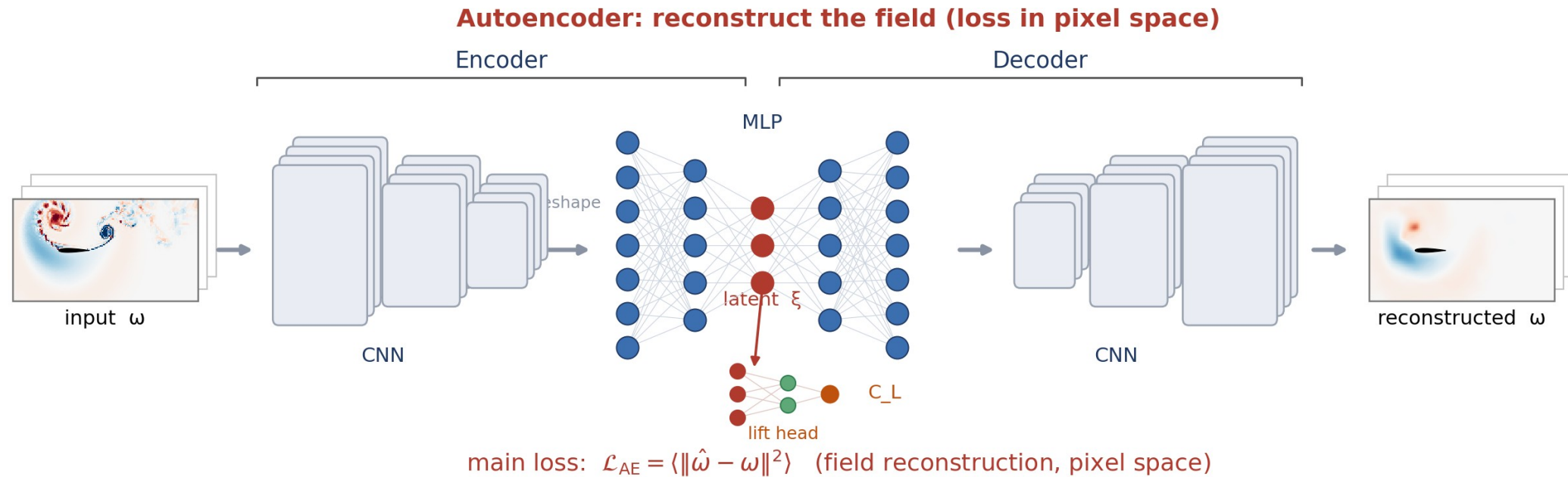
Model detail: encoder and predictor

- **Encoder** = a CNN stem (3 down-sampling stages) + a 6-layer Vision Transformer; the field becomes 288 patch tokens, and a learned [CLS] token is mapped by a small MLP (with BatchNorm) to the latent z ($d = 32 / 64$)
- **Predictor** = a 6-layer autoregressive transformer (hidden 384, 16 heads, dropout 0.1): it reads $z(t)$ and emits the next latent $z(t+1)$. **Unconditioned: no gust parameters enter the predictor**
- **Causal mask**: each step sees only past steps, never the future; **RoPE** encodes the frame ordering so the transformer knows what is earlier or later
- **Training**: match the next latent (teacher forcing) + **scheduled sampling** (feed the model its own predictions over an 8-step rollout, so it tolerates its own errors) + **SIGReg** anti-collapse (keeps z informative without a decoder); AdamW, bf16, 20k steps
- **Architecture inspired by the JEPA world-model line**: I-JEPA / V-JEPA, and the from-pixels LeWM (Maes et al. 2026) and LeJEPA (Balestriero and LeCun 2025)



Illustrated alternative: the autoencoder route

- Reconstructive route (Fukami-Taira lineage): a **CNN encoder**, an **MLP latent ξ** (which also feeds a lift head to C_L), and a **CNN decoder**; the loss compares the reconstructed field to the input in **pixel space**

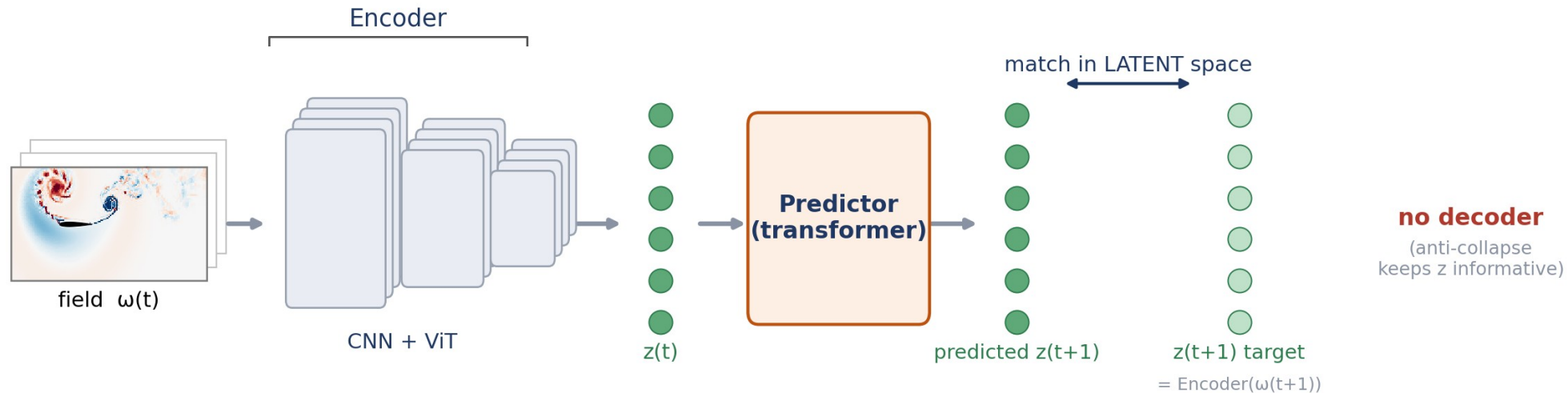


An autoencoder reconstructs the field; the latent is shaped by pixel reconstruction.

Illustrated alternative: the JEPA route

- Same encoder, but a **predictor** advances the latent and the loss lives in **latent space**, with **no decoder** (an anti-collapse term keeps z informative)

JEPA: predict the next latent in latent space, no decoder

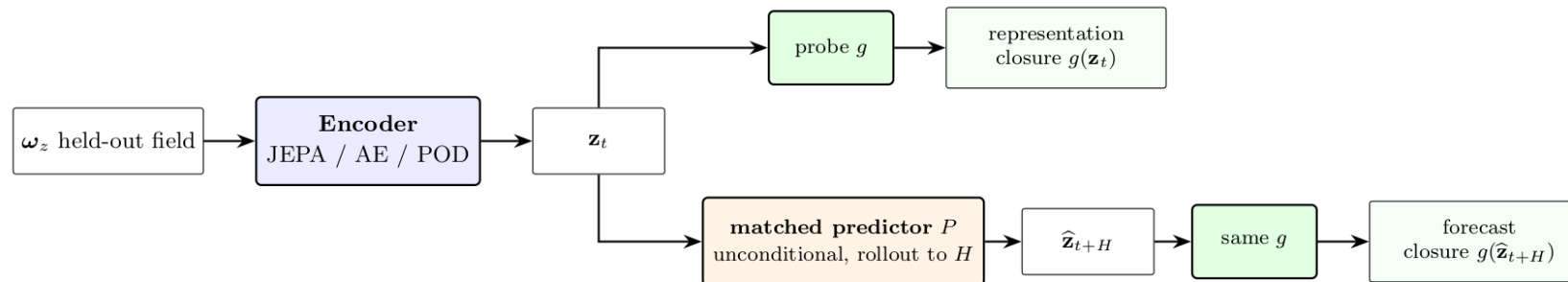


$$\text{main loss: } \mathcal{L}_{\text{JEPA}} = \langle \|\hat{z}_{t+1} - z_{t+1}\|^2 \rangle \text{ (latent space)} + \text{anti-collapse (SIGReg)}$$

JEPA predicts the next latent in latent space, with no field reconstruction.

Closure protocol

- **Representational closure (the headline):** probe each observable from the **encoded** latent at impact + 16, what the state *carries*, no rollout
- **Forward (rollout) mode:** roll the predictor recursively to **H = 16** and probe the predicted latent; here we use it **qualitatively** (does the rolled latent track the wake?)
- **Matched protocol:** same predictor architecture and same probe family, **trained / fitted separately per latent**, so differences are the encoder's
- **Parameter-only floor:** can the **three gust numbers (G, D, Y) alone** predict each observable (a kernel-ridge fit)? Beating the floor proves the latent carries flow state beyond the parameters

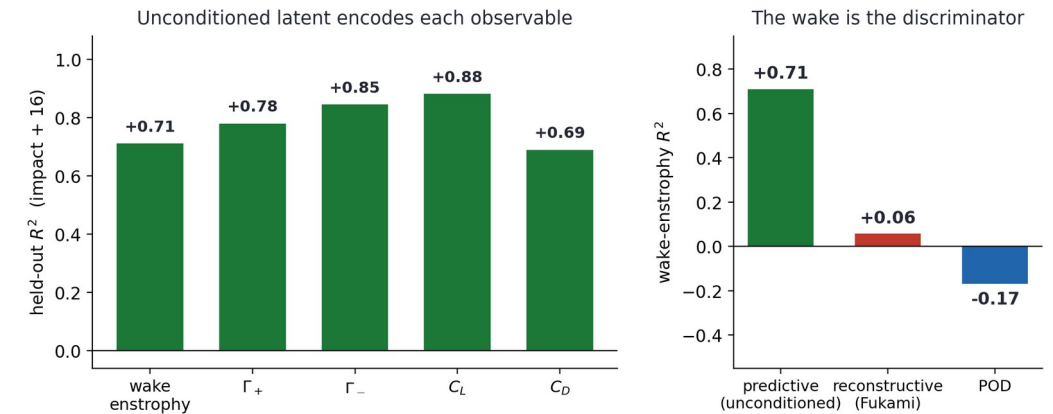


The predictor (unconditional, no gust parameters) and probe family g use the same architecture and protocol, trained separately for each encoder family.

Same probe family maps every encoder (JEPA / AE / POD) to the observables. Representational mode (encoded latent) is the headline; forward rollout is shown qualitatively.

Main result: the latent keeps the wake

- **Representational closure:** probe the **encoded** latent (no rollout) for each observable at impact + 16; the question is what the unconditioned state *carries*
- **Wake enstrophy is the discriminator:** **tf-no-c $R^2 = 0.71$** (the conditioned model 0.75; reconstructive and POD far lower, 0.06 and below)
- So the **unconditioned** latent (no gust parameters anywhere) **encodes the wake that reconstruction smooths away**
- Forces and circulations are read off cleanly too (C_L 0.88, $\Gamma_{\pm}$ 0.85 / 0.78, C_D 0.69)
- **Wake enstrophy** $E_w = \int \omega_z^2$ over the wake ($x/c \in [0.5, 4]$, $|y/c| \leq 1$): the **intensity of the wake's rotational structures** (LEV + shed vortices) that set the future load, the part reconstruction-only states lose

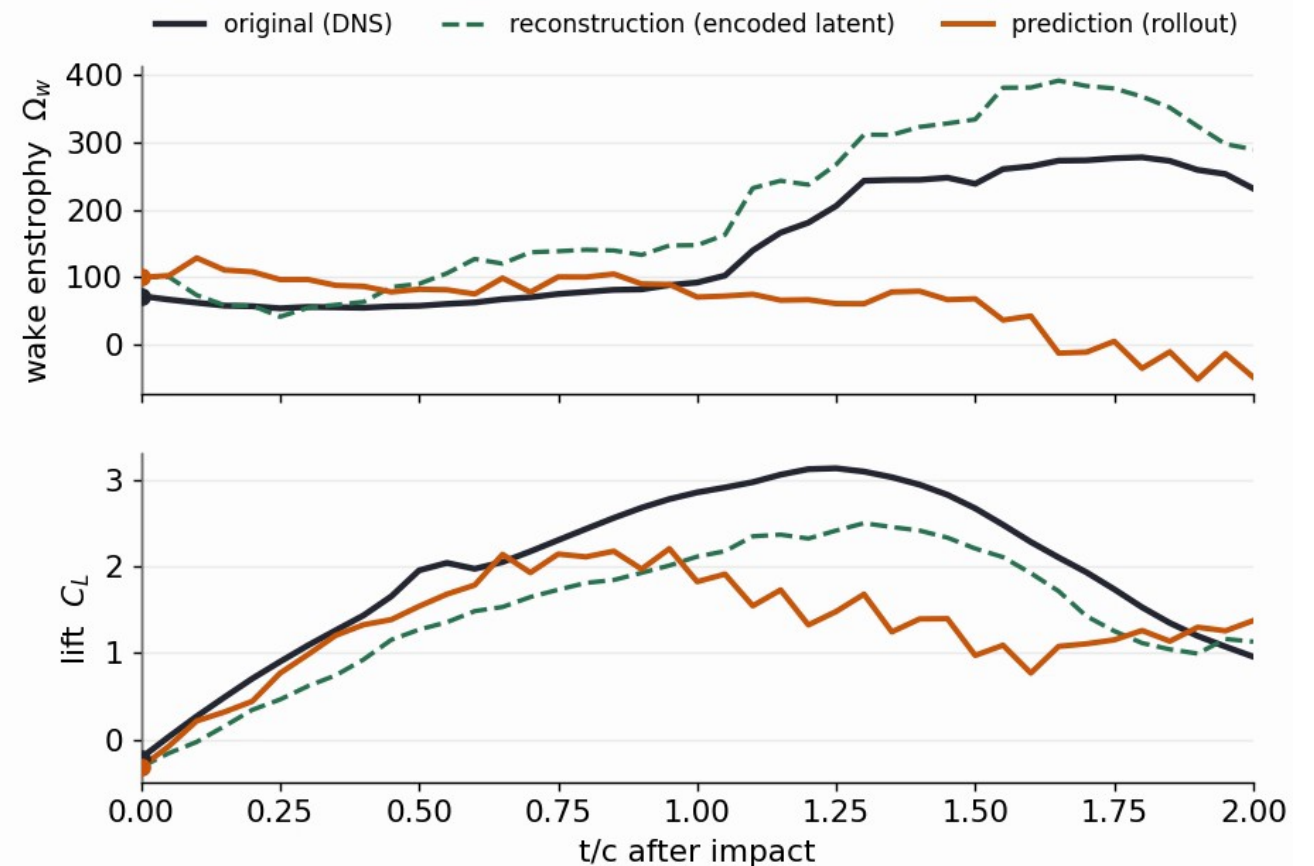


Held-out representational R^2 at impact + 16 from the encoded latent (test_b). Left: per observable (tf-no-c). Right: wake-enstrophy R^2 across families.

The rolled latent qualitatively tracks the wake

- The predictor **rolls the latent forward from impact**; each scalar is read off the rolled latent by a fixed linear probe
- **Wake enstrophy** and **lift** qualitatively track the DNS truth (black) through impact and the lift dip
- **Green = reconstruction** = the model's **own encoded latent decoded with NO rollout** (encode-then-decode of the true frame, the representational **ceiling**); it is **not** a separate reconstructive autoencoder
- **Orange = prediction** = the **rolled latent** (the predictor advanced forward, then decoded)
- A representative **low-error** encounter ($G = +1.5$), not the hardest case; shown qualitatively, no closure R^2 claimed here

(unconditioned) forecast, gust encounter ($G=+1.5$, $D=1.5$, $Y=+0.1$): $t/c = 0$

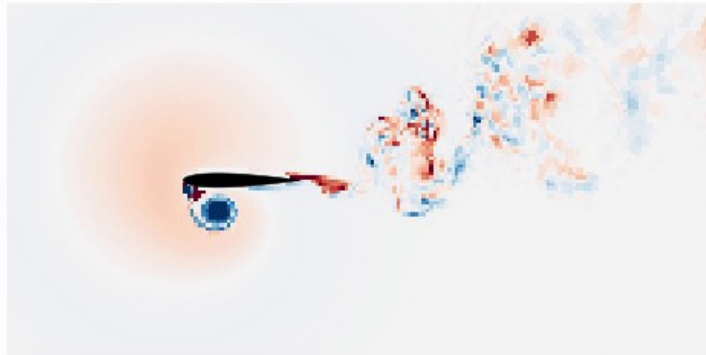


The same rollout in physical space

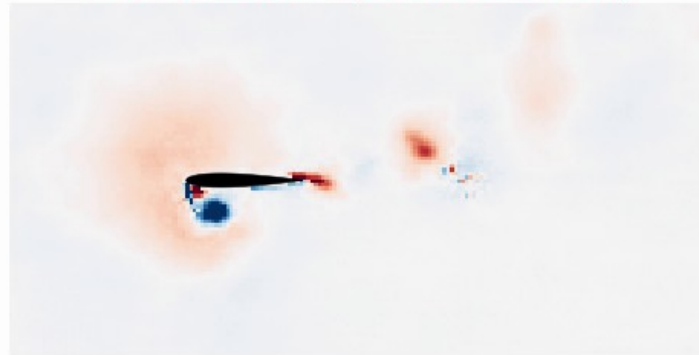
Decode the rolled latent every frame: the rolled latent qualitatively tracks the wake as full vorticity fields, not just scalars.

Mid-plane vorticity (unconditioned), gust ($G=+1.5$, $D=1.5$, $Y=+0.1$): $t/c = 0.00$ after impact

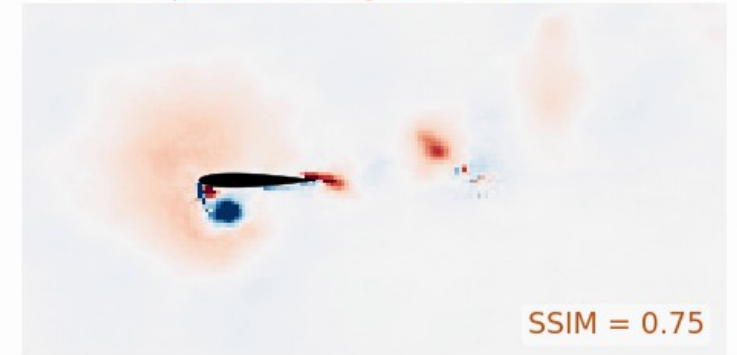
DNS truth



reconstruction (encoded latent)



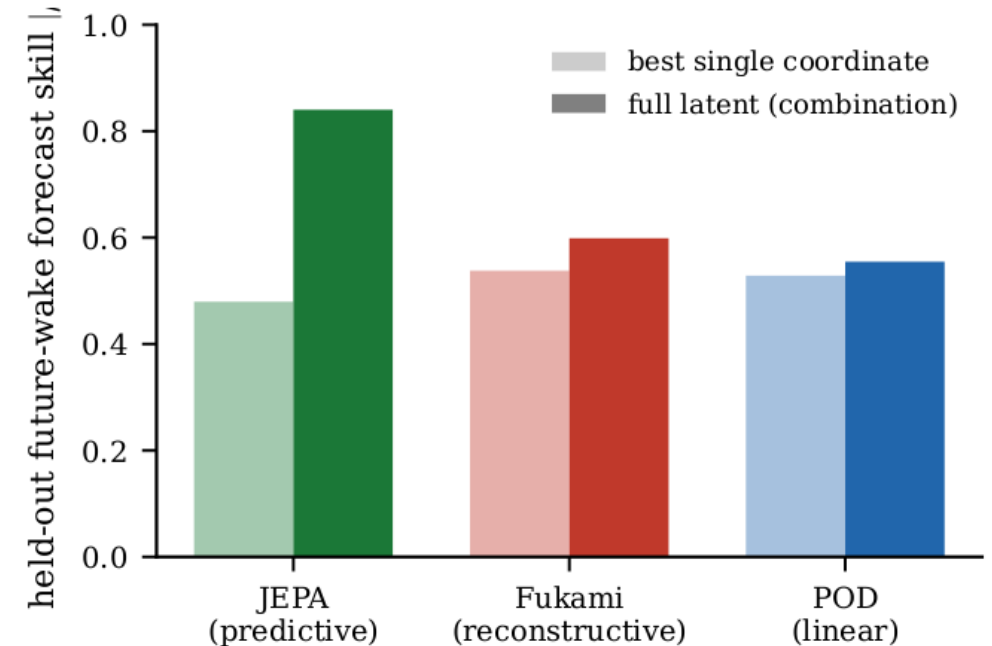
prediction (JEPA rollout)



DNS truth vs reconstruction (JEPA encode→decode of the latent) vs prediction (rolled latent), $G = +1.5$. At impact ($H = 0$) prediction = reconstruction; they diverge with horizon.
SSIM(prediction) 0.75 at impact, 0.63 at $H = 16$. The latent keeps the LEV and shear layer; fine-scale wake is not retained at $d = 64$.

It is the wake, not the forces

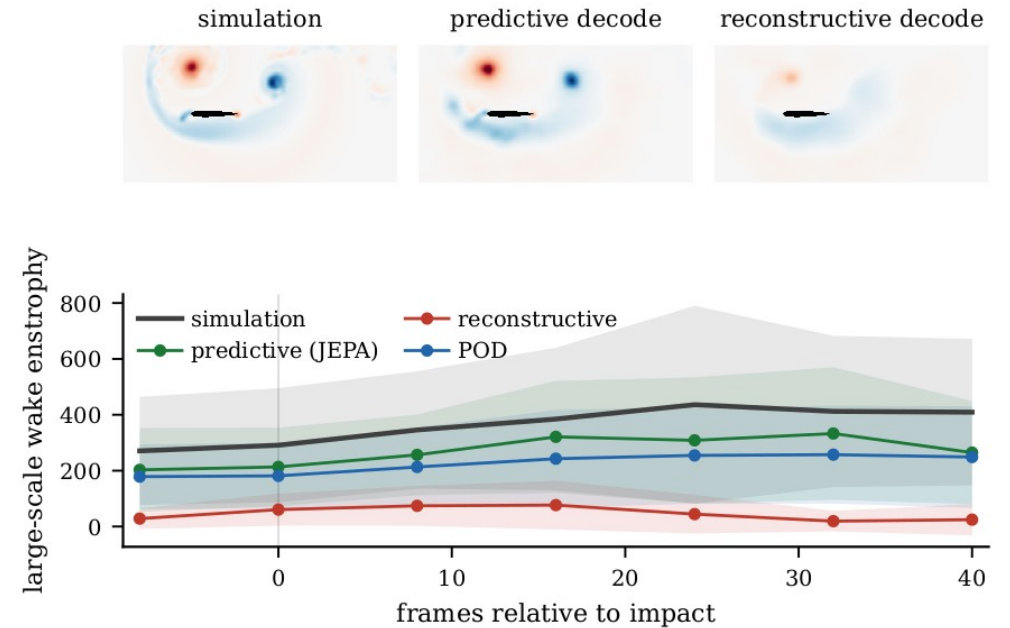
- Each bar is **how well the future wake is forecast**: **dark** = the **full latent** (all 64 coordinates together), **light** = the **single best coordinate**
- **Predictive JEPA**: the full latent (**0.84**) far exceeds the best single coordinate (**0.48**) → the wake is a **distributed code spread across many coordinates**
- **Reconstructive / POD**: the best single coordinate is **already as good as the whole latent** → there is **no distributed / collective wake code** to find
- **Forces** (C_L, C_D), by contrast, are carried **redundantly by every family**, in a single coordinate
- And it clears the **parameter-only floor** → the latent, not the parameters, does the work



Future-wake skill: full latent (dark, all 64 coordinates) vs the single best coordinate (light). Predictive = distributed code; reconstructive / POD = one coordinate suffices.

The wake in physical space: the Gaussian scale split

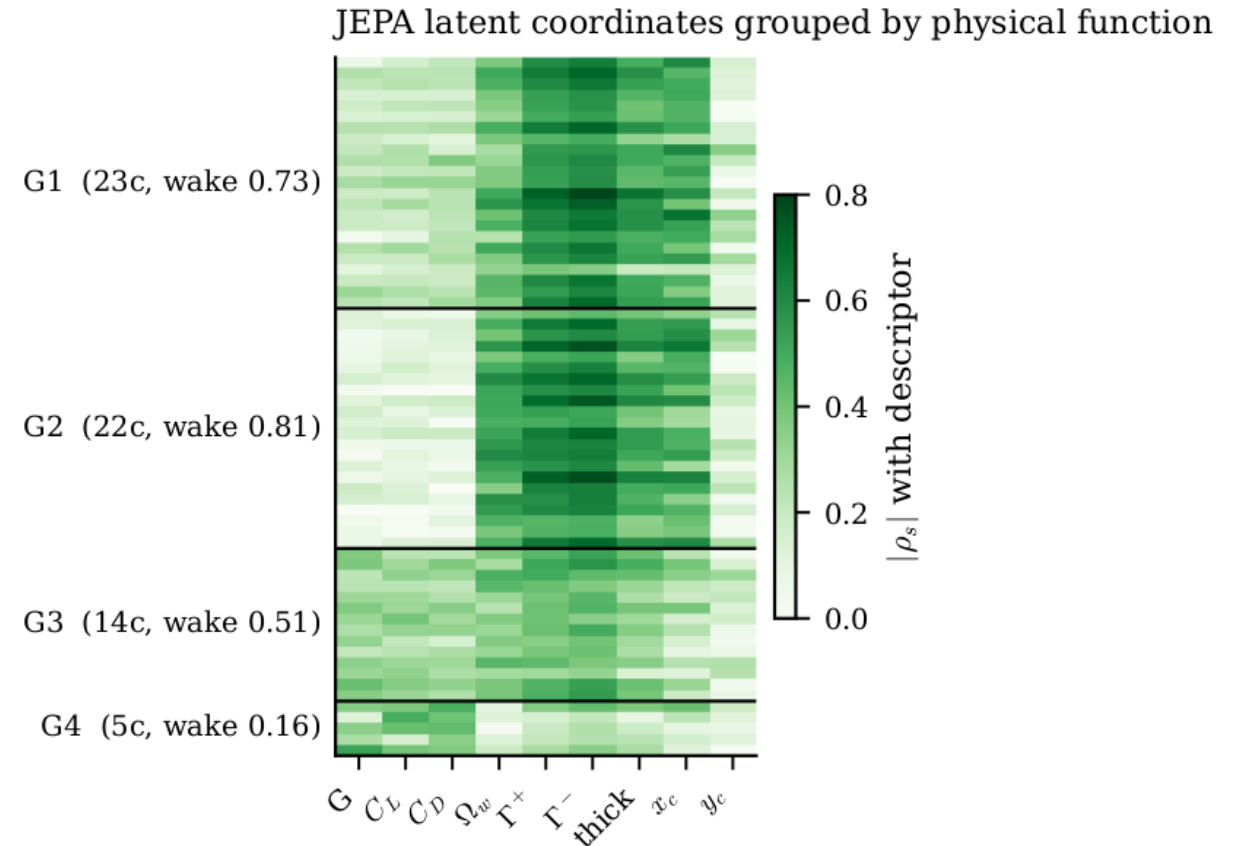
- **Gaussian scale split** (Motoori & Goto 2019): low-pass the vorticity at $\sigma/c = 0.05$ into a **large-scale** part (LEV + shear layer, carries the lift) and a fine part
- Top: large-scale vorticity at impact+16 for the strongest test gust ($G = -3$, $D = 1.5$, $Y = -0.1$), simulation vs the **predictive** and **reconstructive** encode-then-decode reconstructions: predictive keeps the LEV/wake, reconstructive smooths it
- Bottom: the **large-scale wake enstrophy** through the encounter; bands = mean ± 1 s.d. across test_b encounters
- So 'it is the wake' is visible in **physical space**, not only in the scalar closure



Large-scale ($\sigma/c = 0.05$) wake vorticity at impact+16 (top) and large-scale wake enstrophy vs frame (bottom), by family.

Latent coordinates group by physical function

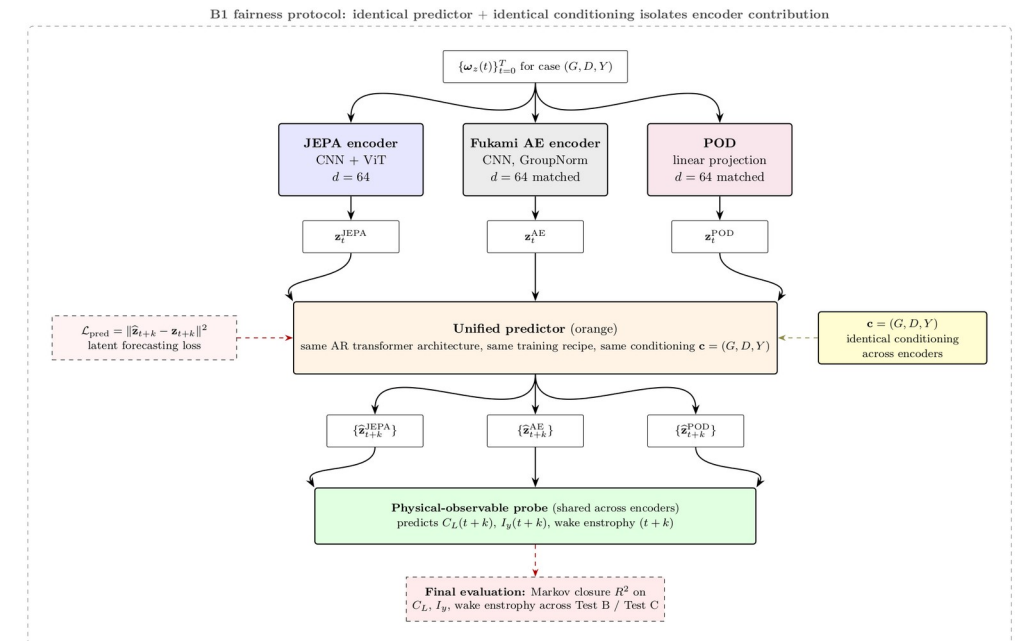
- Profile each of the 64 unconditioned-latent coordinates by its |Spearman| correlation with nine descriptors (gust G , forces, wake enstrophy, circulations $\Gamma_{\pm}$, wake thickness, centroid)
- Clustering the profiles recovers **functional groups: 59 wake-vorticity** coordinates (three wake-dominated clusters) and **5 gust-forcing** coordinates (one force cluster)
- Alone, the wake groups carry the wake at $\approx 0.5-0.8$, the forcing group at **0.16** (all 64 together: 0.84), the collective wake code seen earlier
- Read **descriptively**, not as a causal claim



Per-coordinate $|\rho|$ with each descriptor; rows grouped into functional clusters (G1-G4), labelled with coord count and held-out wake skill.

Controls: objective and supervision, not architecture

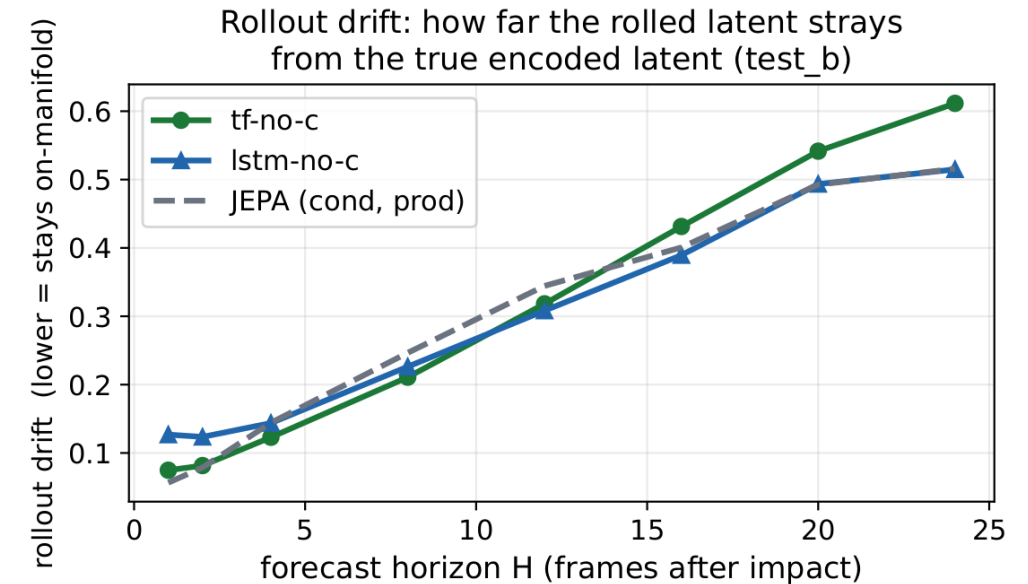
- (Conditioned-model control; no unconditioned variant run.) 2×2 : objective {predictive, reconstructive} $\times$ architecture {CNN, CNN+ViT}, auxiliary heads matched
- **Fair comparison:** every cell is rolled with the **same temporal predictor** (the matched closure predictor: same architecture and protocol, trained separately per family), so the **only thing that varies is the encoder's objective / architecture**
- The **predictive objective wins at both architectures** (wake R^2 0.46 / 0.45 vs 0.16 / 0.29) $\rightarrow$ **it is the objective, not the ViT**
- And the **wake-observable head is a necessary ingredient:** removing it removes the gain
- $\rightarrow$ the gain needs the **predictive objective and wake supervision together**



Objective $\times$ architecture controls, three seeds per cell; every cell rolled with the same matched closure predictor, so only the encoder's objective / architecture varies.

Diagnostic 1: latent drift

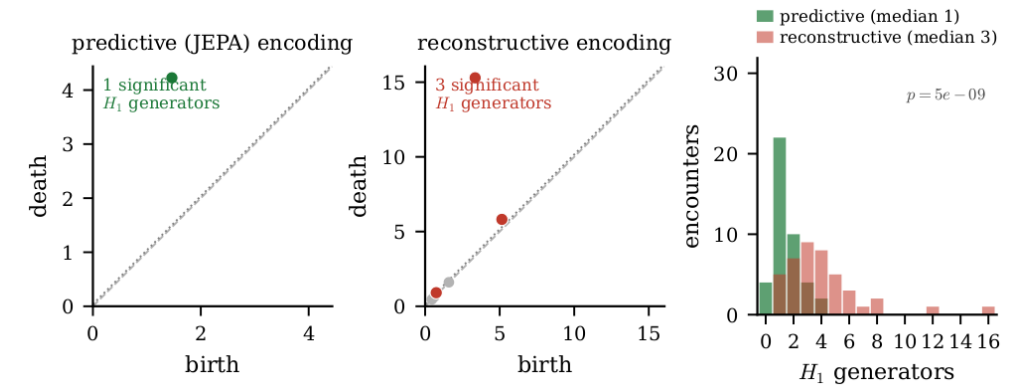
- **Why this question:** a planner / RL agent queries the model at states reached by its **own rollout**, one-step error is not enough
- **z_dns** = the latent **encoded directly from the simulation field** (ground truth);
z_markov = the latent **produced by rolling the predictor forward**
- The curve is their **relative distance vs horizon** ($\|z_{\text{markov}} - z_{\text{dns}}\| / \|z_{\text{dns}}\|$); **lower / flatter is better** (the rollout stays on the data manifold)
- **Result:** removing the gust parameters **does not destabilise the rollout**. Through the **+16** horizon we forecast, the unconditioned latents drift on the **same trajectory as the conditioned model** (≈ 0.4 , within 0.04); **lstm-no-c stays matched to +24** (0.51 vs 0.52), while **tf-no-c** drifts a little more at the long tail (0.61)



Rollout drift vs horizon (test_b): relative distance $\|z_{\text{markov}} - z_{\text{dns}}\| / \|z_{\text{dns}}\|$. The drift grows steadily; the unconditioned families track the conditioned model through the +16 forecast horizon.

Diagnostic 2: topology of the encounter

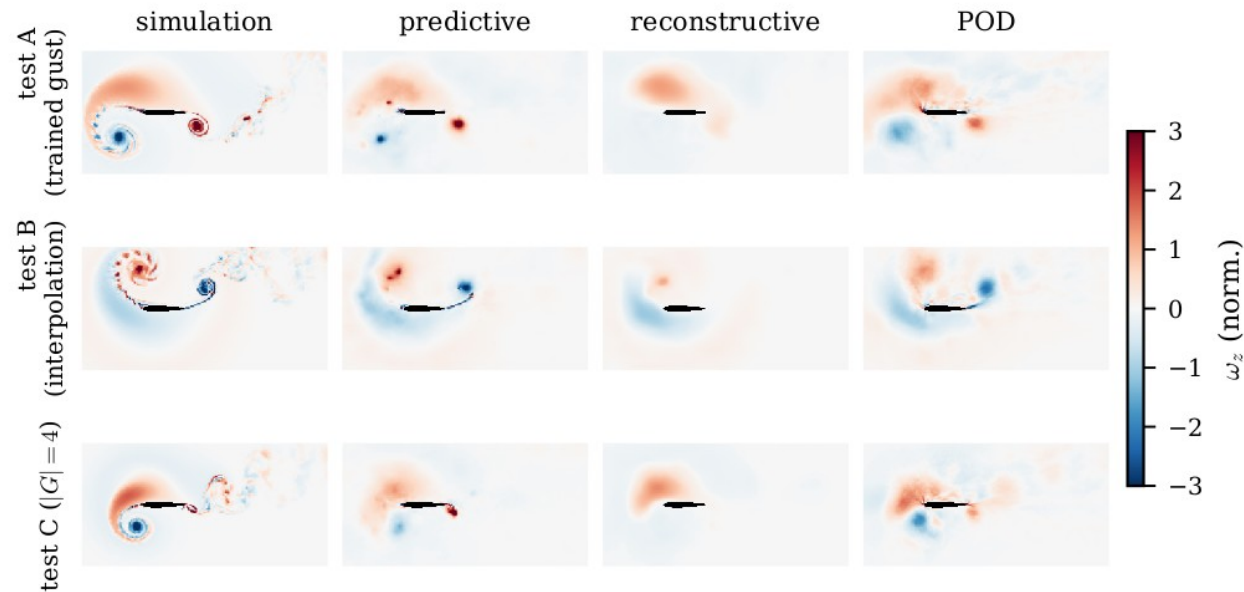
- A gust encounter is a **cycle**: the flow loops and roughly returns to where it started
- We ask whether the latent traces **one clean loop** or a **tangle of many**
- **Persistent homology** is just a principled **loop-counter** on the latent path (it counts the loops that genuinely persist, not noise)
- **Result:** the **unconditioned** JEPA latent makes **one loop** (median 1); the **reconstructive** latent **fragments into many** (median ~ 4); $p \approx 5 \times 10^{-9}$
- So the predictive state captures the encounter as a **single coherent cycle** (count of persistent H_1 generators; Mann-Whitney test)



The latent path of an encounter: the predictive latent traces one loop (median 1), the reconstructive latent many (median ~ 4). Persistent homology counts the loops.

Decoded reconstructions

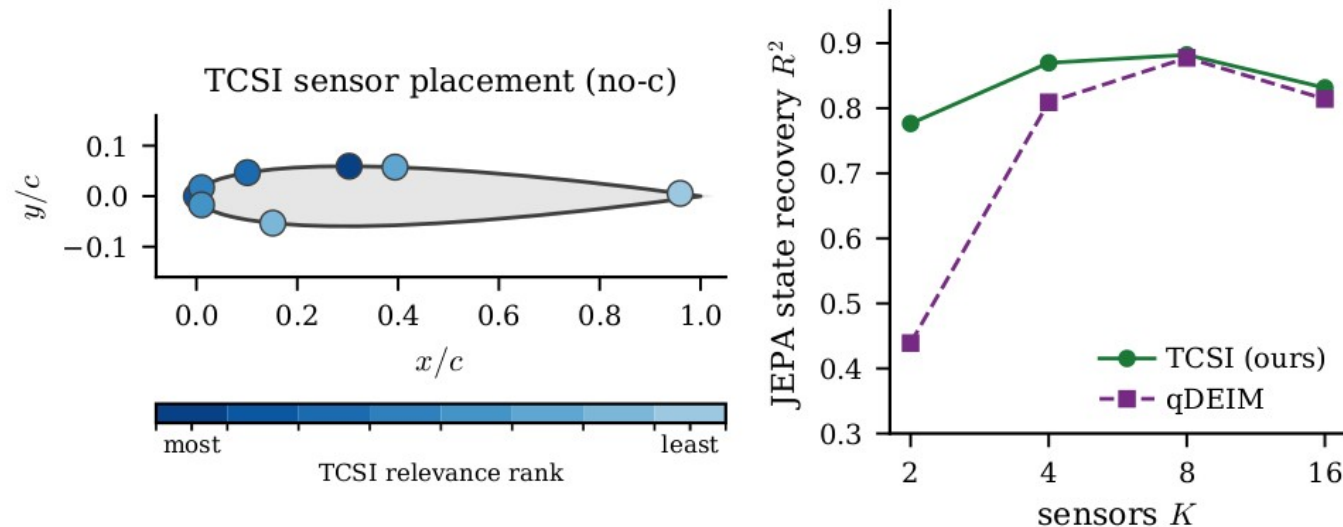
- Decoded fields: predictive vs reconstructive vs POD vs DNS
- The predictive decode is **blurrier pixel-wise** but preserves the **transported large-scale wake**; the reconstructive decode is sharp yet drifts under rollout



Held-out reconstructions; the predictive objective trades pixel sharpness for transported structure.

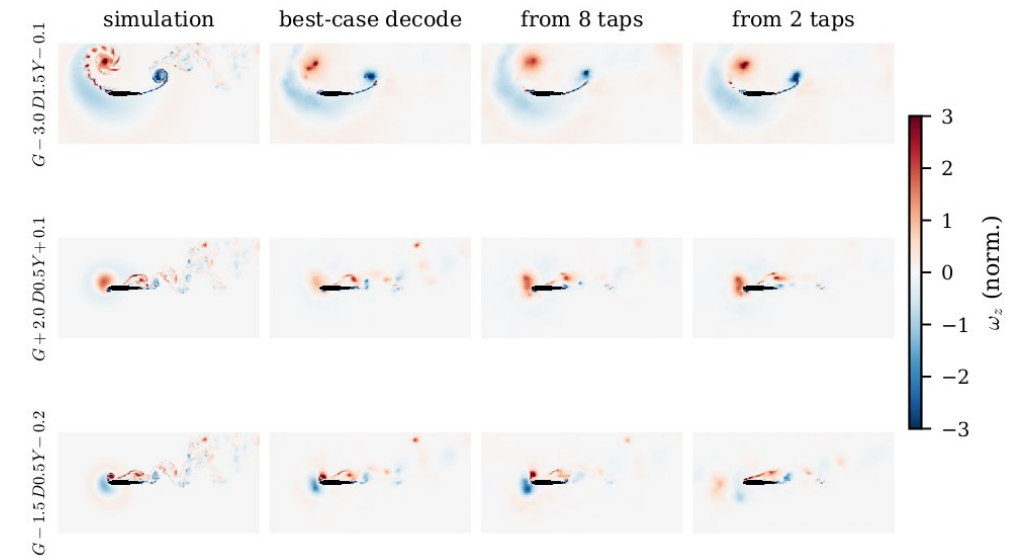
Sparse sensor placement

- Sparse wall-pressure sensors selected by **TCSI / qDEIM** vs uniform
- **Model:** the pressure $\rightarrow$ latent map is a **KernelRidge (RBF)** regressor on the K -sensor $\times$ pre-impact-window vector; $K = 2 / 4 / 8 / 16$ (LSTM / KRR 5-fold-CV selected to guard small-sample overfitting)
- **Metric — state-recovery R^2 :** the **fraction of the held-out latent's variance recovered from the K pressure taps** (1 = perfect, 0 = no better than the mean), for the **unconditioned tf-no-c latent** (no gust parameters anywhere)



Flow recovered from sparse wall pressure

- **Model:** a **KernelRidge (RBF kernel)** regressor maps the K wall-pressure taps over a pre-impact window to the **impact-frame latent**; the frozen decoder then renders the field
- **$K = 8$ taps** recover the leading-edge vortex and shear layer; **$K = 2$** coarsens but keeps the gross wake structure
- Benchmarked against the **best-case decode** (decode of the **true simulation latent**, i.e. the ceiling decode) and the simulation
- So the predictive state is **reconstructible from a few wall sensors**, a deployment-relevant observability result

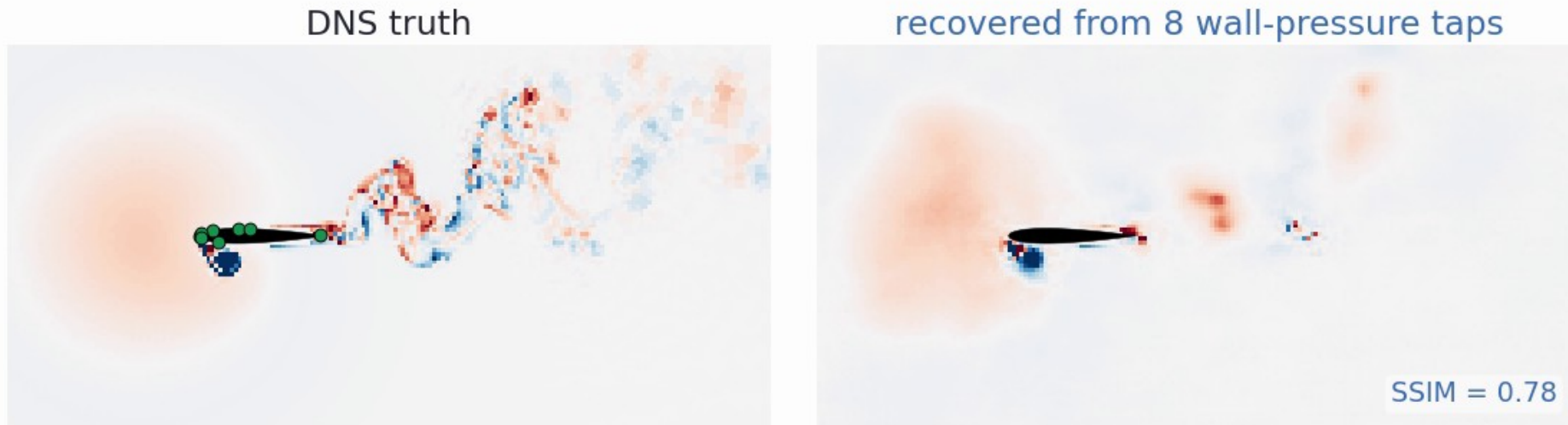


Flow recovered from sparse wall pressure: simulation, best-case (ceiling) decode of the true simulation latent, and decode of the latent estimated from $K = 8$ and $K = 2$ taps (held-out).

Sparse-sensor state estimation in action

No predictor here: a per-frame **MLP** maps a causal 6-frame window of 8 wall-pressure taps to the latent, then the frozen decoder renders the field.

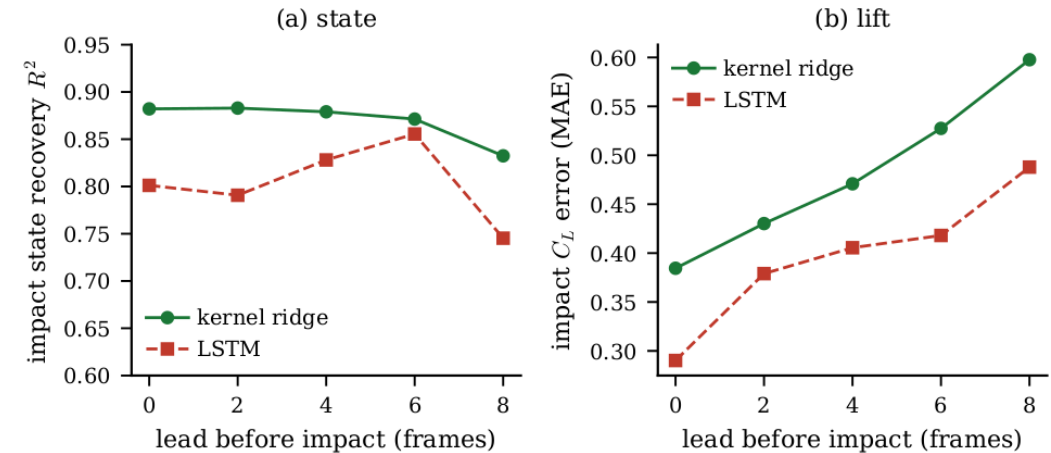
State estimate (unconditioned) from 8 wall-pressure taps: $t/c = -0.40$ before impact



DNS truth (green dots mark the 8 TCSI taps) vs the field recovered from wall pressure alone, on a held-out encounter. Held-out-encounter latent $R^2 = 0.74$, SSIM 0.63. Wall pressure fixes the near-body LEV and shear layer; the far wake is not surface-observable.

Control relevance: observable ahead of impact

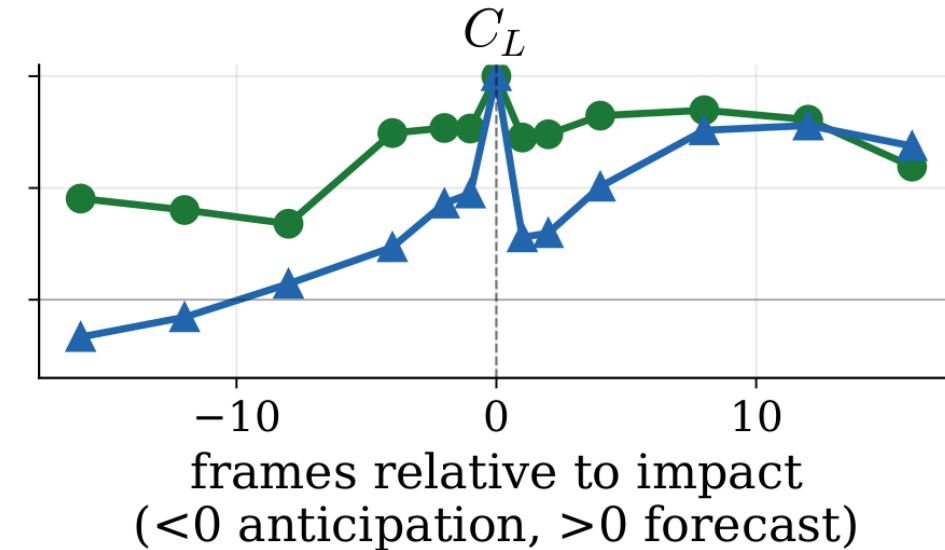
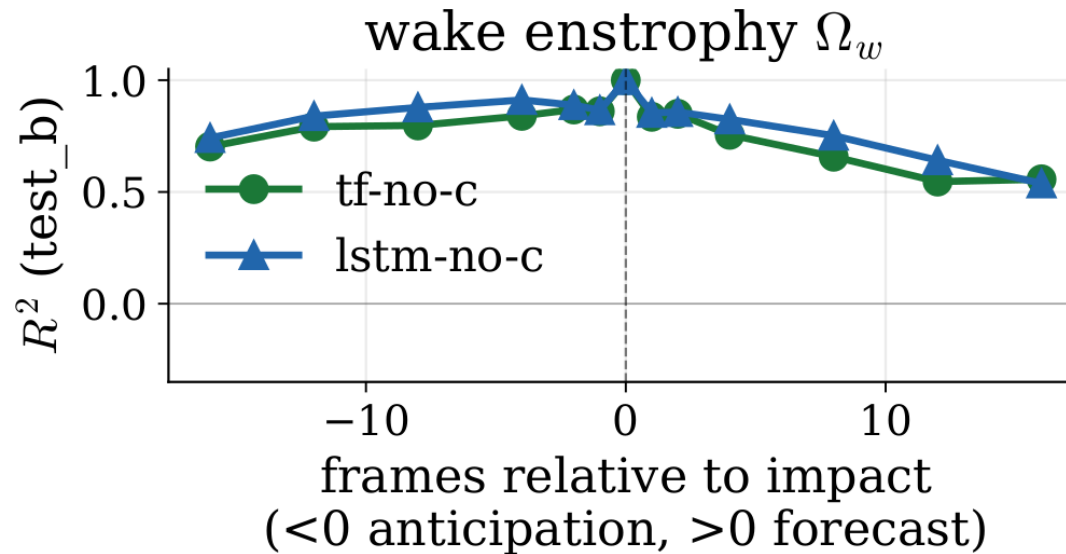
- From a causal window of sparse wall pressure, recover the **impact-frame latent** at a **lead time before impact** (held-out test_b)
- Impact state** is recovered at $R^2 \approx 0.88$ right at impact and stays **above 0.83** out to **8 frames ahead** (kernel ridge; LSTM comparable)
- Impact lift** C_L is estimated with MAE rising gracefully from ≈ 0.38 at impact to ≈ 0.60 at **8 frames ahead** (LSTM lower, 0.29 to 0.49)
- So the **unconditioned** state is **observable ahead of impact** from a few wall sensors: the ingredient a controller consumes



Recovered from sparse wall pressure at each lead time before impact (test_b): (a) impact-state R^2 ; (b) impact- C_L MAE. Kernel ridge vs LSTM.

Two forecast windows: early warning, then forecast

- We roll the unconditioned JEPA's own predictor forward in latent space and read each observable off the rolled latent with a **fixed linear probe**: we predict the JEPA latent, then probe wake / lift from it
- **Wake**: a wide ± 16 -frame ($\approx \pm 0.8$ t/c) window around impact (R^2 0.7-0.9) -- the structural signature gives **early warning** and short-range forecast
- **Lift C_L** : hard to anticipate (flat before impact) but **strongly forecastable after** ($R^2 \approx 0.8$ to +12) -- the load is predictable **once impact hits**



Wake (left) and lift (right) R^2 vs frames relative to impact (test_b): $x < 0$ anticipation lead, $x > 0$ forecast horizon; tf-no-c and lstm-no-c.

Conclusions

- A **fully unconditioned predictive latent** (no gust parameters anywhere) **representationally keeps the wake** (wake $R^2 \approx 0.71$) that **force-only and reconstruction-only** states lose
- That latent **traces a single clean cycle**, is **transport-consistent** within an encounter, and is **observable from sparse wall pressure**
 - **compact** ($d = 32 \approx d = 64$, participation ratio ≈ 1.7)
- The **rollout** is shown **qualitatively**: it stays on-manifold and tracks the wake, but we make **no forward-closure R^2 claim**
- It is a **substrate** for model-based control and RL world-models, **not yet a validated controller**
- **Next**: add an **actuation channel** (same machinery), **close the loop**, push to **3D observability** at $|G| = 4$, and carry the recipe to **other parametric flows**

THANKS

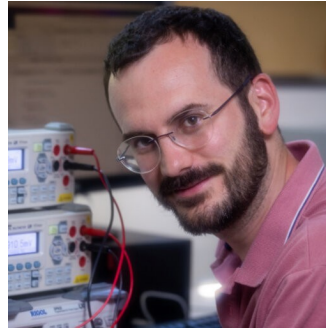
Acknowledgements

This work is a collaboration with:



Alberto Solera-Rico

INTA · UC3M



Arnau Miró

UPC · BSC



Oriol Lehmkuhl

BSC

Work produced with the support of a 2024 Leonardo Grant for Researchers and Cultural Creators, BBVA Foundation, project PREVENT (grant LEO24-2-15988). The Foundation takes no responsibility for the opinions, statements and contents of this project, which are entirely the responsibility of its authors.

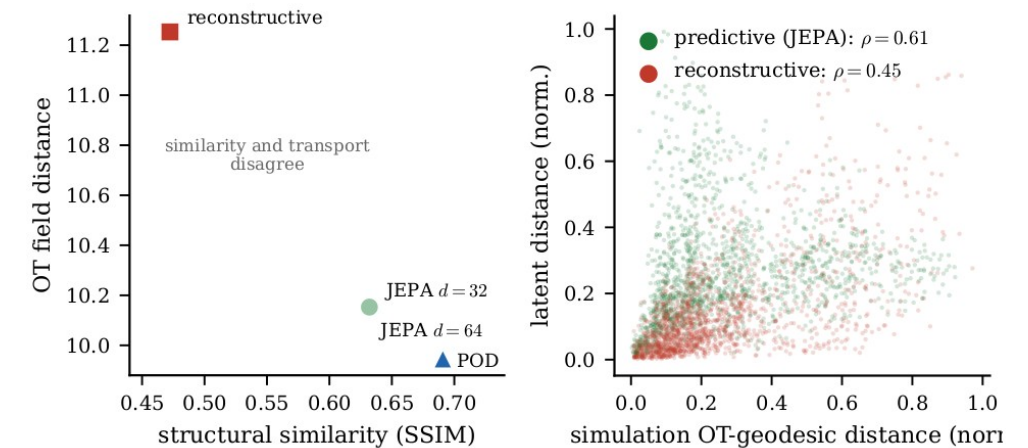


Backup slides

Supporting detail and ablations

Transport geometry (optimal transport)

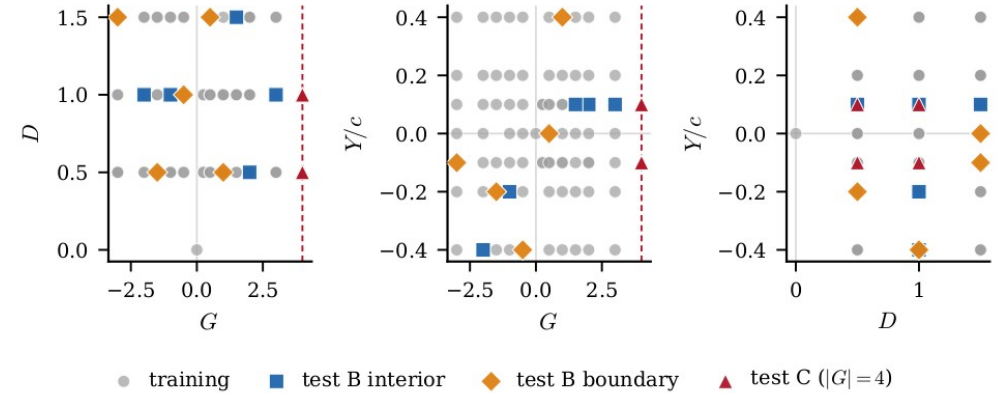
- **Why:** Euclidean / pixel distance ignores **where structures move**; control cares about advection of the LEV and shear layer
- **OT field distance** (after Tran, Yeh & Taira, JFM 2026): split vorticity into $\pm$ parts, transport each with **unbalanced Sinkhorn** OT and sum, the least work to rearrange one field into another
- We did **not** train the latent to match OT (that paper does); this is a **post-hoc test**: per encounter, Spearman-correlate the **latent** distance matrix with the **OT** matrix ($n = 42$)
- **Within-encounter:** the unconditioned JEPA **0.61** vs reconstructive **0.45**, order-preservation along the trajectory, **not** an isometry
- **Honest caveat:** *pooled across* encounters the alignment **does not hold**, so it is **trajectory-local**, not a global latent metric



Per-encounter latent-vs-OT distance alignment (Shepard). OT field distance after Tran, Yeh & Taira (JFM 2026). Within-encounter Spearman 0.61 vs 0.45; pooled it does not hold.

Dataset and protocol

- Partition **v2** (locked, sha256-anchored): 84 cases
- **226** train encounters · **42** test_b · **24** test_c ($|G| = 4$)
- ω_z pipeline v1: spatial mask + p99.99 clip + 3σ scale (train_std 3.55), no mean shift (preserves vorticity antisymmetry)
- Reporting: bootstrap $n = 2000$ CIs · 3 encoder seeds · 5-fold probe CV



Training envelope in (G, D, Y) .

Matched-capacity: $d = 32$ vs $d = 64$

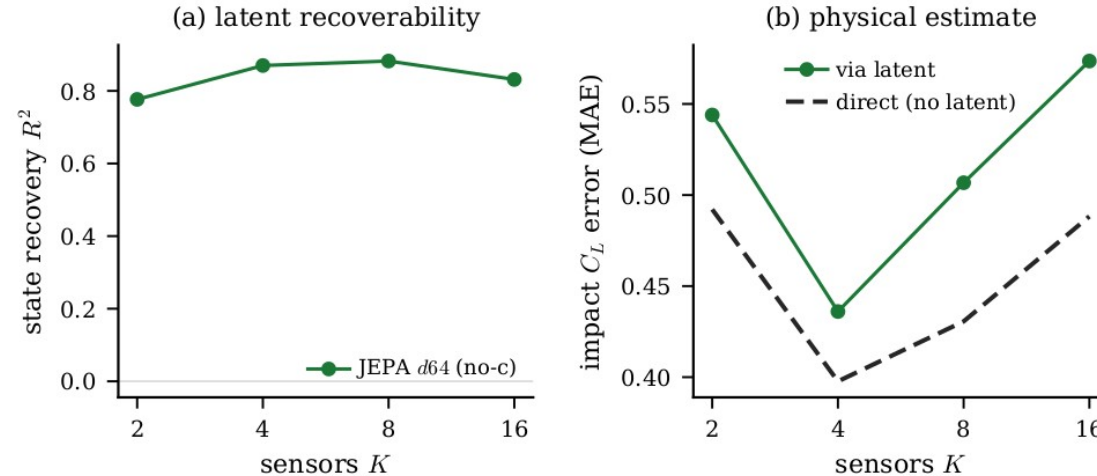
- (Conditioned-model control; no unconditioned variant run.) Halving the latent leaves the representation and **every mechanism diagnostic** intact
 - representational wake R^2 0.74 ($d=32$) vs 0.75 ($d=64$); drift ratio 0.86 vs 0.85; OT 0.61 vs 0.63
- Cost is **in-distribution forecast sharpness**: $H = 16$ wake closure 0.45 $\rightarrow$ 0.21
- Participation ratio ≈ 1.7 , leading PC $\approx \frac{3}{4}$ of variance $\rightarrow$ the effective dimension is a handful

Seed variance

- **(Conditioned-model control; no unconditioned variant run.)** Three encoder seeds per cell; wake-closure means reported $\pm$ standard deviation
- The reconstructive CNN+ViT cell has **large seed variance (± 0.27)**, consistent with the drift mechanism: without a forward-predictable geometry, closure is unstable across initialisations
- Predictive cells are tight (± 0.03 – 0.06)

Parameter observability from the latent

- $z \rightarrow (G, D, Y)$ probe R^2 from the rolled-out latent, $K = 8$ pre-impact sensors
- test_b: **G 0.46 • D 0.80 • Y 0.10**, gust strength and core diameter recoverable; cross-stream offset Y is marginal
- test_c ($|G| = 4$): negative, the 3D observability boundary of a single mid-plane slice



Gust parameters implicit in the latent on held-out cases.

SSIM convention

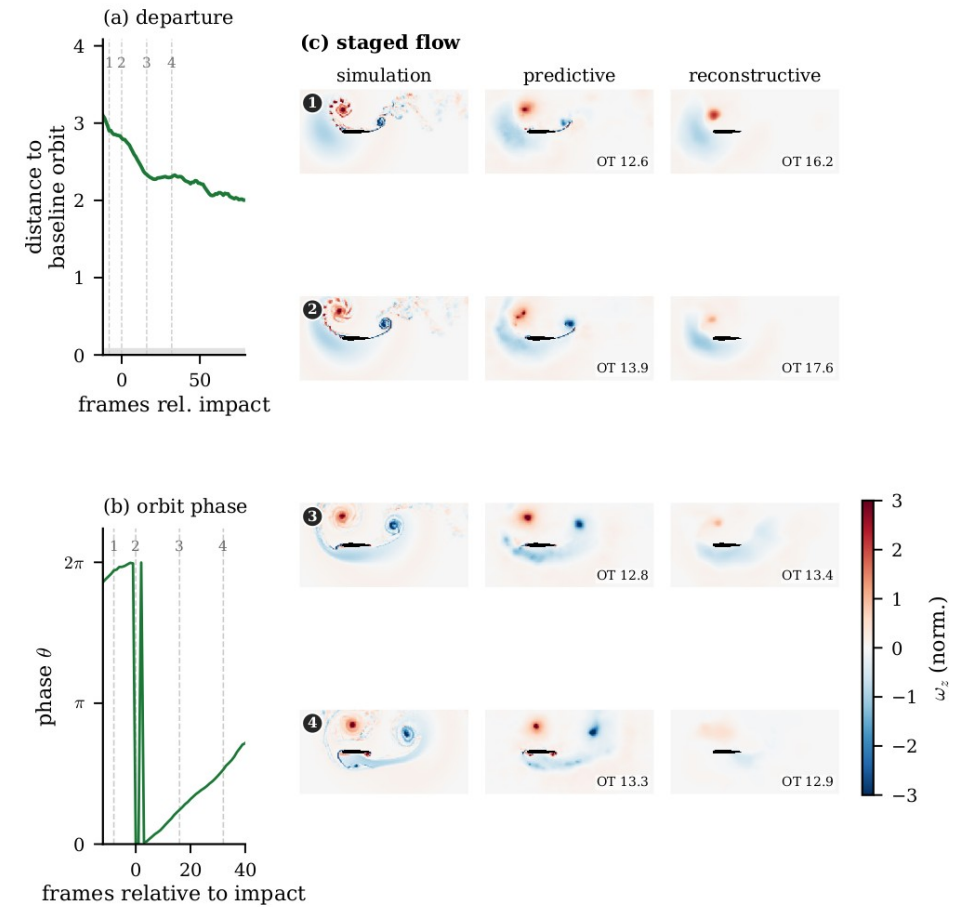
- Reconstruction SSIM uses the **Wang convention** ($K1 = 0.01$, $K2 = 0.03$) on pipeline-normalised ω
- Data range **$L = 2 \cdot \text{global p99.9}(|\text{target}|) \approx 8.31$** (split v2)
- Decoder test_a SSIM $\approx$ **0.73** at this convention

Parameter-only floor

- KRR-RBF on (G, D, Y) → observable at impact; train / test_b / test_c R^2
 - on train, (G, D, Y) interpolates C_L and C_D well (226 points in 3-D)
- On **test_b** the floor **collapses** across observables; on **test_c** it is negative for 5 of 6
- → JEPA's generalisation is **not** explained by the gust parameters alone

Phase-amplitude reading

- Phase-amplitude decomposition of the encounter cycle in the predictive latent
- The sharp $2\pi \rightarrow 0$ step in panel (b) is a **phase wrap** (θ is a cyclic angle on the orbit; it resets at the branch cut, which falls near impact), **not a physical jump**
- Connects the predictor to the **sensitivity-function control** designed for these flows
- An actuation channel is the natural next input for closed-loop control



The $2\pi \rightarrow 0$ step in panel (b) is a phase wrap (cyclic angle resetting at the branch cut near impact), not a physical discontinuity.